Guarded fixed-momentum tracking on a public NSLS-II design model

Technical review packet | 23 October 2026, America/New_York | Independent research prototype

Measured result: 1.61x lower first-workload time versus the fastest tested existing method that passed the frozen accuracy gate: **2.676 s to 1.665 s**. Tracking alone improves by 2.05x. The new-method cost includes plugin build, lattice construction, compilation, warm-up and guard-triggered native reruns.

The workload and the accuracy gate

The source is BNL-authored NSLS2/PyELEGANT demo/nsls2cb65pm.lte, pinned to commit 11d97113ea7ae05fd0ffe710a88793f2bdc9ed84. It is a historical complex-bend design demonstrator: 3 GeV, 791.958 m, 3,030 elements. RF, radiation, collective effects and magnet errors are off; no mechanical apertures are present. This is not the present operational machine.

A frozen independent-seed confirmation compared 15 predeclared variants on 64 launches for 256 turns. Survival labels must match wherever native 40- and 80-step references agree. Sixteen core launches must have maximum position error at most 1 micrometre and momentum error at most 1 microradian-equivalent normalized momentum. No threshold was relaxed after seeing results.

The selected fused40_s4 passed all 63 reference-agreed labels. Point 48 remains unresolved because the references disagree. Core errors were 0.368 micrometres and 0.0317 microradian-equivalent. All 25 guard reruns per repetition are charged. This is empirical qualification on this finite test, not a uniform stability theorem.

Method	Qualified	First workload, s
native10	no	6.709
native40	yes	25.343
analytic10	yes	2.676
fused40_s10	yes	2.205
fused40_s2	yes	1.907
fused40_s4	yes	1.665

The roughly 15.2x ratio versus native40 is not the headline: built-in analytic passes already supply much of that improvement. The additional measured gain over analytic10 is 1.61x. No tenfold industrial tuning advantage is established.

Primary context: BNL-230151-2026-TECH reports nearly 30,000 HSR field-error simulations; BNL already develops nonlinear-optics and data-driven optimization methods. This contribution claims a qualified implementation benchmark, not invention of matrix transport or accelerator optimization.

Sources: github.com/NSLS2/PyELEGANT; technotes.bnl.gov/Home/ViewTechNote/230151; bnl.gov/nsls2/accelerator/research.php. Full bibliographic and provenance records accompany the package.

Why this reduction is mathematically admissible

At fixed momentum deviation, eligible drifts, pure quadrupoles and linear-field bends are affine in the four transverse coordinates in the selected PyAT paraxial model. We compose the actual discrete integration matrices, retaining thick sextupoles. The selected method preserves 40-step linear bodies and uses four sextupole steps. A fused C pass combines each matrix with an input-domain guard. Guard rejection means uncertainty and triggers a native rerun; it is never silently classified as physical loss.

Independent C-formula review and 108 element/momentum cases with 100 points each found maximum transverse residual 6.94e-18. A separate fused-kernel check passed 100 accepted points and three guard failures. Prefix bounds use a large amplitude margin. They are not outward IEEE error certificates. Floating regrouping can affect chaotic trajectories, so element-level identities do not replace the held-out test.

Longitudinal path length is not preserved. A conservative exact-rational magnitude argument bounds omitted linear contributions below 10^60 over 4,096 turns; retained nonlinear finiteness and complete IEEE equivalence remain separate. The method requires no RF or other longitudinal-to-transverse feedback. It is not a six-dimensional tracker replacement.

Costs, reproduction and limits

Reference generation cost 75.00 s; all 15 confirmation methods, setups, repeats and one build cost 164.77 s. Earlier exploration is additional. Conditional repayment needs 58 future equal-cost batches for references alone, or 126 additional batches for the measured campaign. This assumes reuse without further validation costs. Identical inputs can be cached. No operating-machine or end-to-end optimizer saving has been measured.

The LTE requests N_KICKS=40, but the PyAT importer defaults to NumIntSteps=10. Explicit step policies and refinement are therefore recorded. Matching step counts does not prove ELEGANT equivalence; no native-ELEGANT cross-code validation was available. All failed variants and the unresolved reference point are retained.

A longer post-selection backcheck used 128 launches and 512 turns. The selected compiler met the same gate; a subsequently added analytic20 comparator gave an observed 3.86x first-workload ratio, with only one baseline timing. This is not independent confirmation. Further single-point diagnostics were nonmonotone: one boundary launch survived 40/80 steps, was lost at 160, and survived at 320. Fine-pair agreement is therefore not a proof of physical convergence; these boundary cases remain refinement-sensitive.

The package includes original licensed model, portable C/header sources, frozen protocols and executed sources, raw arrays, independent audits and offline replay. From research/bnl_tracking, install requirements in an isolated Python environment; use build_kernel.py to build with GCC, check_results.py for offline audit, and replay_confirmation.py for a new live run. Recorded outputs are immutable.

Requested independent evaluation

A useful next assessment is a BNL-maintained lattice and accuracy contract, including relevant apertures, momentum offsets, nonlinear errors and long-horizon requirements. Test the native analytic-pass workflow as well as the compiler. Adoption should follow a measured complete-cost advantage under that contract; this packet does not recommend connecting the prototype to machine controls.